Discover: The Moon's Journey

Today I can… ☐ State that the Moon orbits Earth in roughly a month. ☐ Explain that moonlight is reflected sunlight. ☐ Explain phases using our changing view of the lit half.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Lesson strip

Arrival

I can explain why the Moon appears to change shape as it orbits Earth.

In your book

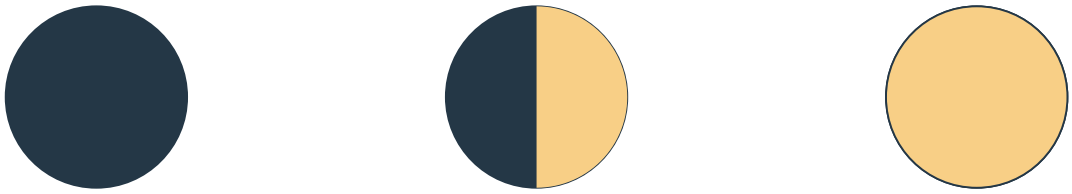
In your book: Glue in your strip; answer four prompts.

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Arrival · Supported



1.

Point to the full Moon picture.

2.

Link to today: Choose whether planets move around the Sun or Earth.

3.

Choose: sunlight or the Moon’s own light?

4.

Point to the part of the lit Moon you can see.

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Exit Ticket · Supported

1.

Does the Moon make its own visible light?

In your book: Under the heading Exit ticket: answer the prompt; show one next step.

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Arrival · Standard

1.

Name one useful feature and one limit of an orbit model.

2.

What force keeps planets in orbit?

3.

Explain where moonlight comes from.

4.

Predict why the visible lit part changes.

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Exit Ticket · Standard

1.

Explain why the Moon appears to change shape.

In your book: Under the heading Exit ticket: answer the prompt; show one next step.

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Arrival · Stretch

1.

You move a model by hand. Why does this not measure gravity?

2.

How does an orbit differ from a spin, or rotation?

3.

Use the Sun, Moon and the person who looks. Explain how reflected light goes from one to the next.

4.

Think of a normal phase and an eclipse. What difference would you expect?

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Exit Ticket · Stretch

1.

Why are ordinary phases not caused by Earth’s shadow?

In your book: Under the heading Exit ticket: answer the prompt; show one next step.

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Starter and method strip

Starter

Look at new, quarter and full Moon. What might be changing?

Method strip

1. Keep the cool light still. Use one Moon model or the phase cards you have.
2. Mark the lit half. Mark where you look from. Stop and check with the adult: which way does the light come from?
3. Put four views in order. Look from the same place each time. Explain a phase and an eclipse on their own.
4. In your book, label what you saw. Say why the lit part you can see changes. Put the lights back as they were. Return the kit.

Safe access

Cool LED torch, large foam ball, observer marker and four phase cards.

Adult uses a low-power cool light; no lasers or looking at the Sun. No blackout: normal-light diagrams are equal routes. Keep the ball above the observer’s head shadow so a false eclipse is not created. Agree before any light change; adult handles equipment and cutting. Keep movement clear.

Only the apparatus needed for the current stage is on the table.

If you cannot safely take part, watch the teacher demonstration and record the same observation and conclusion in your book.

After the task, put what you saw in your book. Say what it shows about today's goal.

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You Do · Independent

Independent

Make or sequence a four-position phase strip or wheel. Mark sunlight, the lit half and the Earth-view phase. Explain an eclipse separately.

Put four model views in order. Say why the lit part we see changes.

I · S · C

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You Do · Supported

Supported

Make or sequence a four-position phase strip or wheel. Mark sunlight, the lit half and the Earth-view phase. Explain an eclipse separately.

Match new, quarter, full and quarter views, on your own or with a partner.

Use a fixed sunlight arrow and point to the lit half.

The Moon appears… because we see… as it orbits…

I · S · C

Discover: The Moon's Journey

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You Do · Choice

Choice

Make or sequence a four-position phase strip or wheel. Mark sunlight, the lit half and the Earth-view phase. Explain an eclipse separately.

Use phase cards to label four model views. Choose each label: full / quarter / new. Order the labelled views around the model. Mark sunlight direction and the lit half. Choose the cause: our changing view of the lit half / the Moon changing shape. Complete: We see ___ because ___. Choose the lunar eclipse cause: Earth blocks sunlight / the Moon becomes smaller. Record C beside your book response.

I · S · C

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You Do · Aspire

Aspire

Make or sequence a four-position phase strip or wheel. Mark sunlight, the lit half and the Earth-view phase. Explain an eclipse separately.

Explain the phases in order. Say how an eclipse differs. Give one limit of the model.

Say when our view changes. Say when light is blocked. Explain the tilt of the orbit.

I · S · C

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Knowledge organiser

Learning goal

I can explain why the Moon appears to change shape as it orbits Earth.

Blue shows word meanings. Green shows key ideas.

Word and meaning

satellite — an object that orbits a planet

phase — the shape of the lit part of the Moon visible from Earth

orbit — a curved path around another body

reflected — light that changes direction from a surface

Key ideas

Torch is the Sun, ball is the Moon, observer is Earth. Keep the sunlight direction fixed.

Half the Moon is illuminated by the Sun. As it orbits, our view of the illuminated half changes.

When the visible lit half faces us, we see full Moon. When it faces away, we see new Moon.

Trace the light path: Sun → Moon → observer on Earth. Name the Sun as light source and the Moon as reflecting object.

Keep the claim fair

The model does not show realistic sizes, distances or the tilt of the Moon’s orbit.

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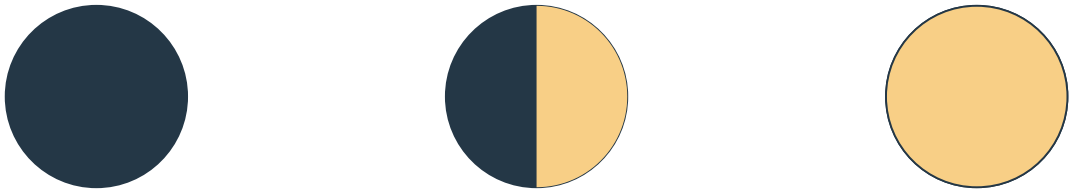
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Knowledge organiser · continued

The Moon's Journey

New, half and full are views from Earth. The Moon stays round; its visible lit part changes.

Orange shows the part seen lit. Dark blue shows the part not seen lit.



new Moon | half Moon | full Moon

Views from Earth; not to scale.

The ball stays round at every position.

Discover: The Moon's Journey

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Word page

The Moon's Journey

Blue shows word meanings.

satellite — an object that orbits a planet

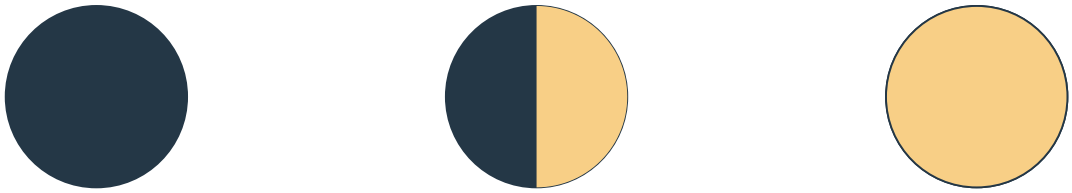
phase — the shape of the lit part of the Moon visible from Earth

orbit — a curved path around another body

reflected — light that changes direction from a surface

New, half and full are views from Earth. The Moon stays round; its visible lit part changes.

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new Moon | half Moon | full Moon

Views from Earth; not to scale.

The ball stays round at every position.

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We Do · response cards

Response cards

A B C

We Do

What normally causes Moon phases?

Our changing view of the sunlit half

Earth’s shadow every month

The Moon making its own light

Discover: The Moon's Journey

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We Do · cards and recording

We Do

Choose the destination for each card. Explain the evidence clue.

Read the card.

Choose its destination.

Check the reason and repair.

Keep one explanation in your book.

Crescent

Only a small part of the illuminated half faces us.

Full Moon

The lit side faces the place we look from.

New Moon

The lit half faces mostly away from the observer.

Destinations

Ordinary phase

Lunar eclipse

Incorrect claim

Guided example

In your book: Record one placement and its science reason.

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We Do · cards and recording

We Do

Choose the destination for each card. Explain the evidence clue.

Read the card.

Choose its destination.

Check the reason and repair.

Keep one explanation in your book.

Dark red Moon

Earth blocks direct sunlight during this event.

Own light

The Moon glows by itself.

Every month

Every full Moon must be an eclipse.

Destinations

Ordinary phase

Lunar eclipse

Incorrect claim

Guided example

In your book: Record one placement and its science reason.

Discover: The Moon's Journey

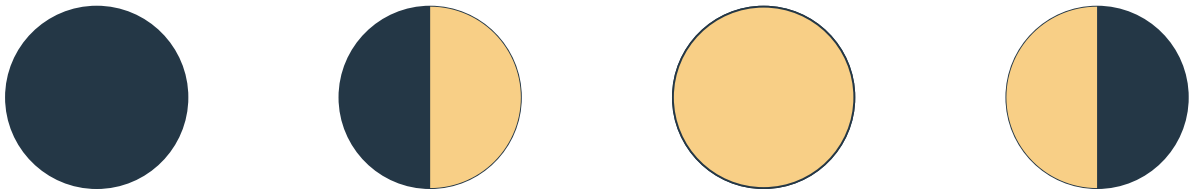
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You Do · phase cards

Phase cards

Make or sequence a four-position phase strip or wheel. Mark sunlight, the lit half and the Earth-view phase. Explain an eclipse separately.



new quarter full quarter

Sunlight direction →

Method strip

Keep the cool light still. Use one Moon model or the phase cards you have.

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